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Abstract: In this paper, the process and structure of discharge in a long rod-plane gap have been clarified further by the aid of Lichtenberg figure technique. A new model for simulating the effects of the positive space charge has been established. With the help of the new model, the electric field distribution along the axis of a rod-plane gap during the discharge is studied quantitatively. On the basis of the new model, the mathematical models of calculating the inception voltage (V_{cs}), the length (L_{cs}) and the charge quantity (Q_{cs}) of the positive corona streamer are used. The calculated values are in good agreement with experimental values.

Introduction

When a high voltage switching surge of positive polarity is applied to a long rod-plane air gap, the discharge develops through the following main phases[1]: the first corona; the leader channel with the leader corona streamer (simply called as corona streamer) in front of its tip; the final jump and the return stroke. The most important ionization phenomena in breakdown process occur in the front of the corona streamer region, supported by the electric field created by the leader channel (or electrode) and the positive space charge remained in the corona streamer region. The former can be calculated by using a numerical method, the latter can be determined by a space charge model, and is also one of the most important factors influencing the characteristics of discharge.

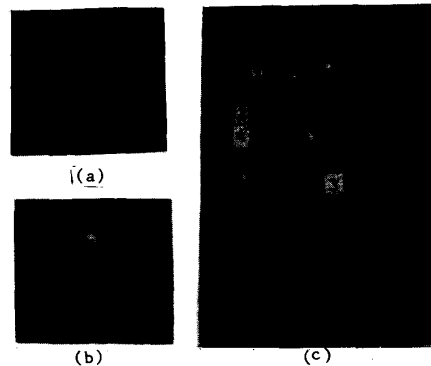
Previous works on the effects of the positive space charge produced by corona streamer growth in non-uniform field system has been concerned with the experimental determination of their electrical properties and the formulation of a quantitative model. In 1965, Dawson and Winn[2] proposed a model to simulate the effect of positive space charge by using point charge representing the charge in the streamer tip. This model has successfully elucidated the possibility propagating even in zero ambient field. The model has been developed more rigorously by Gallimberti in 1972[3] through an equivalent electron avalanche and an energy balance condition. Phelps[4] has indicated that positive space charge is deposited on the streamer channel, except when the streamer reaches the cathode. In 1974, Khaled[5] introduced a new approach to simulate the effect of the space charge by use of ring charges. In 1976, Arima and Watanabe[6] introduced a model in which space charge produced by corona streamer distribute within a part of sphere of radius being the corona streamer length. But there exist two shortcomings. The first one is the formula of space charge density assumed is not reasonable at some points; the second one is that the method [6] they used to calculate applied field needs to be improved.

In this paper, the authors have introduced a new model which can overcome these two shortcomings, on the basis of the model proposed by Arima and Watanabe. With the help of the new model, the discharge models for simulating the discharge characteristics of corona streamer are used. The results calculated have been shown a very good agreement with the values of experiments.

Experimental results using Lichtenberg figure technique

Figure 1 shows typical Lichtenberg figures for the application of a positive switching surge. Referring

to Figure 1(a) and (b), the typical Lichtenberg figures for corona streamer phase can be seen clearly. For close examination of the figures, it suggested that the weak luminous corona streamers extend from the rod tip within about a part of sphere. Figure 1(b) is the result of twice-applied voltages, which shows quite similar to that of Figure 1(a).



(a) $U=16.8$ kv, $T_f=100$ μ s, $R=0.1$ cm, $D=100$ cm;
(b) the same as (a) except for twice-applied voltages;
(c) $U=158.3$ kv, $T_f=100$ μ s, $R=0.1$ cm, $D=80$ cm.

where U : an applied voltage;
 T_f : a wavefront of the voltage;
 R : the radius of the hemispherical of a rod-plane air gap;
 D : the gap length.

Figure 1. Typical discharge patterns obtained by using Lichtenberg figure technique.

Referring to Figure 1(c), a typical Lichtenberg figure for a streamer-leader system can be seen. For more detailed examination of the figure, one can see that the Lichtenberg figure is a total record of the discharge pattern for different phases except the final jump and breakdown. The first corona streamers, the corona streamers before a leader inception, a leader channel and the corona streamers in front of the leader tip are indicated by No.1, 2, 3, 4, respectively. It suggests that with the increase of an applied voltage, the first corona streamers are initiated and extended from the rod tip. Afterward, a dark period may exist. When corona streamers develop efficiently, a strongly luminous leader extends from the rod tip.

Space charge model

Figure 2 shows a model of the distribution of space charge proposed by Arima and Watanabe[6]. The formula of space charge density in the volume shown in Figure 2 was given,

$$\rho(r) = A/r^B \quad (1)$$

where $\rho(r)$ is defined as the space charge density at a point, which is situated at distance r from the rod tip; A and B are constants

An examination of (1) shows that when r approaches zero, $\rho(r)$ would approach infinity. Physical reality

requires $\rho(r)$ to be finite for all values of r concerned. The experimental result also shows that $\rho(0)$ should be a finite value[7]. Hence a new expression must be given,

$$\rho(r) = A/(r+C)^E \quad (2)$$

where $\rho(r)$, A and B are the same as (1); C is constant.

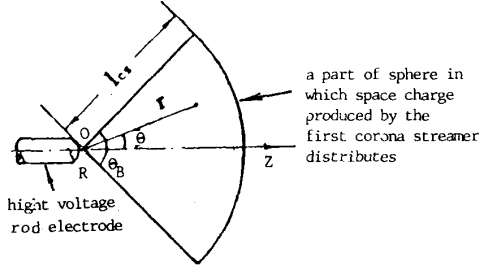


Figure 2. The distribution of space charge

Electric Field Analysis in Long Rod-Plane Air Gap Discharge

During the entire evolution of the streamer-leader system, we make the following assumptions: (a) the first corona streamers emanated from the tip of the rod electrode have developed in the space in absence of space charge. The electric field distribution only depends on applied voltage and electrode geometry, which is called applied field or external field; (b) After the initiation of the first corona streamers, the corona streamers extended from the tip of the rod electrode (or a leader) are called the second, third... corona streamer. They develop in the electric field which has two components— one due to the applied (or external) field and the other, due to the space charges produced by the first corona streamer.

Suppose that E indicates the electric field at arbitrary point on Z axis coinciding with the symmetrical axis of rod-plane gap, then

$$E = E_{ex} + E_{sp} + E_I \quad (3)$$

where E_{ex} is defined as applied field (or external field) calculated by charge simulation method[8]. This method is efficient to be used in the discharge model to calculate the field at the point concerned in the step by step process; E_{sp} is defined as the electrical field created by positive space charge produced by corona streamer and distributed in the volume shown in Figure 2; E_I represents the electric field owing to negative charges of high voltage rod electrode induced by the positive space charges. The analytical expressions can be derived as following:

$$E_{sp} = -\frac{1}{2\epsilon} \int_0^{\theta_B/2} \int_0^{l_{cs}} \frac{\rho(r) r^2 \sin\theta}{r_1^3} (r \cos\theta - Z) dr d\theta \quad (4)$$

$$E_I = -\frac{1}{2\epsilon} \int_0^{\theta_B/2} \int_0^{l_{cs}} \frac{R r^2 \rho(r) \sin\theta \cos\theta_4}{r_2^3} dr d\theta \quad (5)$$

where $r_1 = \sqrt{(r \cos\theta - Z)^2 + (r \sin\theta)^2}$; $r_2 = \sqrt{(R + r \cos\theta)^2 + (r \sin\theta)^2}$; $r_3 = R^2/r_2$; $\theta_3 = \sin^{-1}(r \sin\theta/r_2)$; $r_4 = \sqrt{(Z + R - r \cos\theta)^2 + (r \sin\theta)^2}$; $\theta_4 = \sin^{-1}(r \sin\theta/r_4)$;

$\theta_4 = \sin^{-1}(r_3 \sin\theta_3/r_4)$; ϵ is a dielectric constant in air; θ_B is an angle shown in Figure 2; l_{cs} is corona streamer length from the tip of rod electrode (or a leader); R is the radius of the hemispherical tip of the rod electrode; $\rho(r)$ is given in (2).

Experimental results[1] have concluded that: (a) the electric field in the corona streamer region is about 5 kv/cm, i.e. $E = 5$ kv/cm; (b) The electric field distribution in the corona streamer region is nearly uniform. The constant A is chosen to satisfy (a); B and C are chosen to meet (b). Usually, B is 1.5 and C is about 0.035. Besides, suppose that Z_m is the drift distance from t_1 to t_2 , then Z_m can be given as follows:

$$Z_m = \int_{t_1}^{t_2} b [E_{ex}(Z_c) + E_I(Z_c)] dt \quad (6)$$

where b is the mobility of positive ions ($b = 1.36 \times 10^{-3}$ cm²/kv.us); Z_c is the coordinates of the point at where $E_{sp}(Z_c)$ is equal to zero. Z_c is used to illustrate the drift of the positive space charge distributed in the volume shown in Figure 2.

E_{sp} , E_I and Z_m can be obtained by integrating the formulas of (4), (5), (6) by Gaussian numerical integration.

The Calculated Results and Discussion

The calculation of electric field distribution during gas discharge

Figure 3 shows the electric field distribution in the vicinity of the rod electrode tip owing to the space charge after the first corona streamer occurrence, which are calculated under the following conditions: for a rod-plane air gap, gap length is 80 cm; a switching surge having the crest value of 320 kv with the time of wavefront of 100 us is applied to the gap. The inception voltage of the first corona streamer is 27.5 kv; the length of the corona streamer development is 2.7 cm. Referring to Figure 3, it can be seen that the electric field due to the positive space charge produced by the first corona streamer reduces the field in the vicinity of the rod electrode tip very sharply and increases the field in the front of the space charge region.

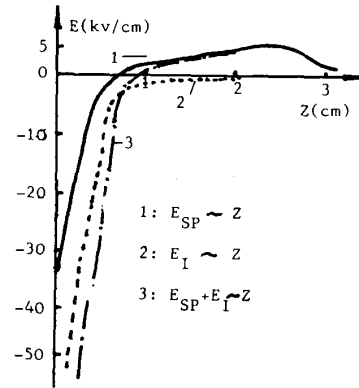


Figure 3. Electric field distribution in the vicinity of the rod tip owing to space charge

Figure 4 shows the effect on the electrical field near the rod tip owing to the drift of the positive space charges. It suggests that a small drift toward the low voltage plane electrode can make a rapid increasing of

the electric field in the surface of the rod tip, so as to that a new corona streamer can incept. The drift of space charge also is one of the most important factors resulting in the occurring of dark period in the discharge process of long air gap:

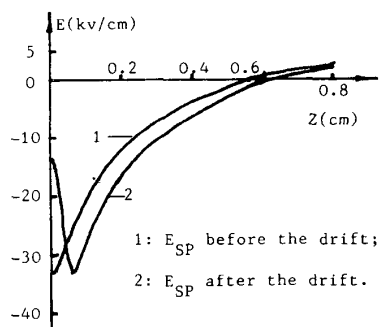


Figure 4. The variation of the electric field due to the drift of positive space charge

Figure 5(a) shows an applied field distribution varying the applied voltage in the vicinity of the rod tip of the high voltage electrode. The applied field very fastly decreases with the increasing of Z being from the tip.

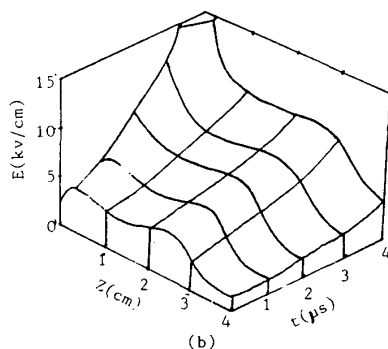
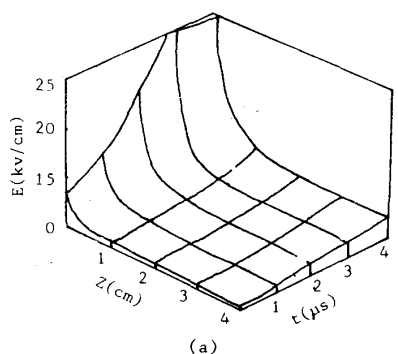


Figure 5. The electric field distribution in the absence and the presence of the positive space charge in the vicinity of the rod tip

Figure 5(b) shows the electric field in the presence of space charge produced by the first corona streamer in the process of gas discharge with the same condition as Figure 4. It can be seen that when the first corona streamer incepts, the electric field in the corona

streamer region is of the range from 4 to 5.5 kv/cm, which has a very good agreement with the experiment value of 5.2 kv/cm[1].

The simulated results of discharge characteristics for a rod-plane gap

On the basis of the aforementioned new space charge model, the mathematical models of calculating the inception voltage, the length and the charge quantity of positive corona streamer were established[9]. The results shown in Figure 6, 7, 8 were obtained on a rod-plane air gap under the following conditions: gap spacing is 80 cm and 40 cm. Wave front duration time of a positive impulse voltage was varied from 15 μs to 600 μs. Figure 6. shows the relation between the inception voltage of corona streamer and the wavefront of an applied voltage (for the gap length 40 cm). Figure 7. shows the length of corona streamer development as a function of the wavefront of an applied voltage (for the gap length of 40 cm). Figure 8. shows the charge quantity of corona streamer as a function of the wavefront of an applied voltage for both gap length of 40 cm and 80 cm. it is suggested that the effect of space charge increases the inception voltage, length and the charge quantity of corona streamer. With the increasing of the wavefront of an applied voltage, the inception voltage, length and this charge quantity of corona streamer decrease. It also can be seen that with the help of a

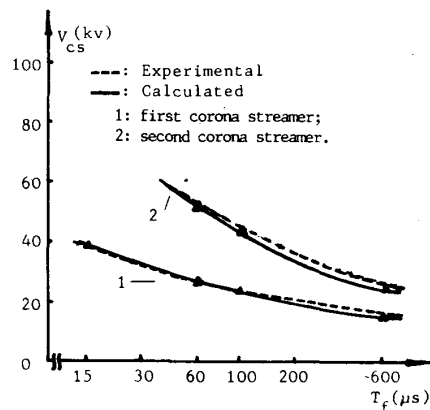


Figure 6. The voltage of corona streamer inception as a function of the wavefront of an applied voltage (for 40 cm gap)

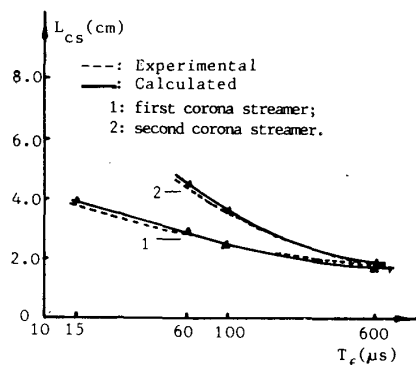


Figure 7. The length of corona streamer development as a function of the wavefront of an applied voltage (40 cm gap)

new space charge model, the simulated results of the discharge characteristics of corona streamer show good agreement with the experimental results[6].

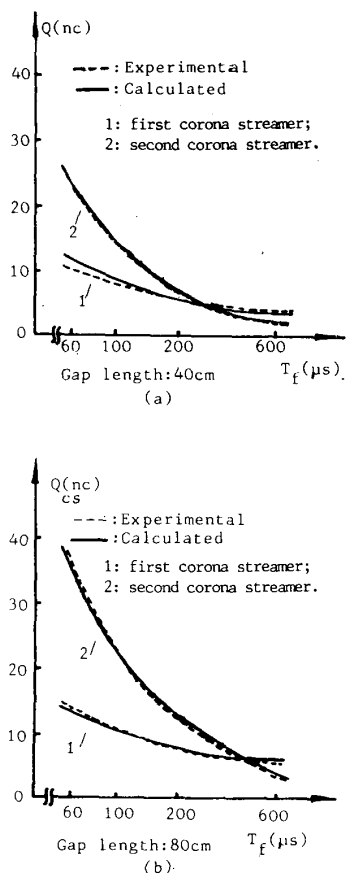


Figure 8. The charge quantity of corona streamer as a function of the wavefront of an applied voltage.

Conclusions

- 1). The new model for simulating the effects of space charge proposed here is reasonable either in the physical picture of the space charge distribution, or the effects on the discharge models quantitatively.
- 2). The effect of the drift of space charge shows the same as that of the distortion of the electric field due to space charge.
- 3). The inception voltage, the length and the charge quantity of corona streamer decrease with the increasing of the wavefront of the applied voltage. The simulated results are in good agreement with the experimental results for the rod-plane gap of 40 cm and 80 cm.

Acknowledgment

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